

Multiple Choice (20 marks)

1. The pressure at the bottom of a pond does NOT depend on the
 - (a) acceleration due to gravity.
 - (b) water density.
 - (c) depth of the pond.
 - (d) surface area of the pond.
2. Heat engines, such as a jet engine, are more efficient when run at
 - (a) high temperatures
 - (b) constant temperatures
 - (c) low temperatures
 - (d) a constant rate
3. An ideal battery of constant voltage V supplies current to a circuit that has total resistance R . If R is doubled, what will happen to the total power...
 - (a) It will double.
 - (b) It will quadruple.
 - (c) It will be halved.
 - (d) It will remain the same.
4. The fact that the Moon always shows its same face to Earth is evidence that the Moon rotates about its axis about once per
 - (a) day
 - (b) month
 - (c) year
 - (d) None of these for the moon does not rotate about an axis
5. How many electrons are there in the third shell of sodium, Na (atomic number 11)?
 - (a) None
 - (b) One
 - (c) Two
 - (d) Three
6. What happens to the pressure, P , of an ideal gas if the temperature is increased by a factor of 2 and the volume is increased by a factor of 8 ?

- (a) P decreases by a factor of 16.
 - (b) P decreases by a factor of 4.
 - (c) P decreases by a factor of 2.
 - (d) P increases by a factor of 4.
7. A pair of protective headphones reduces the sound of a jet engine from 140 dB to 110 dB. What percent of the noise is being blocked?
- (a) 99.90%
 - (b) 78.60%
 - (c) 21.40%
 - (d) 0.10%
8. Heat comes from the Sun to Earth by the process of
- (a) conduction
 - (b) convection
 - (c) radiation
 - (d) All of these, although radiation is greatest
9. The mutual induction of electric and magnetic fields can produce
- (a) light
 - (b) energy
 - (c) Both of these
 - (d) Neither of these
10. Immediately after two separated charged particles are released from rest both increase in speed. The particles therefore have
- (a) the same sign of charge.
 - (b) opposite signs of charge.
 - (c) either the same or opposite signs of charge.
 - (d) Need more information.

True / False (20 marks)

Answer True or False and justify briefly.

1. True or False: The correct answer to 'Big brother and little sister can balance on a seesaw because of balanced' is 'forces'.

2. True or False: The correct answer to 'When you squeeze an air-filled party balloon, you increase its' is 'weight'.
3. True or False: The correct answer to 'Astrophysicists are able to identify the elements in the outer layers of a star by studying its' is 'spectrum.'.
4. True or False: The correct answer to 'The period of a 10-Hz wave is' is '1.0 s'.
5. True or False: The correct answer to 'Any atom that emits an alpha particle or beta particle' is 'becomes an atom of a different element always.'.
6. True or False: The correct answer to 'Consider a ball rolling in a horizontal circular path on the inside surface of a cone. The normal force on the ball' is 'is less than mg always'.
7. True or False: The correct answer to 'A 1134-Hz tuning fork is sounded at the same time a piano note is struck. You hear three beats per second. The frequency of the piano...' is '1134 Hz'.
8. True or False: The correct answer to 'Noise-canceling devices make use of sound' is 'amplification'.
9. True or False: The correct answer to 'Shock waves are produced by waves of sound' is 'All of these'.
10. True or False: The correct answer to 'A barometer that uses water instead of mercury will be' is 'taller'.

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